

Program : Diploma in Chemical Engineering	
Course Code : 3072	Course Title: Mechanical Operations
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L: 2 T: 1 P: 0)	Periods per semester: 45

Course Objectives:

- To impart thorough knowledge in mechanical operations used in chemical plants.
- To impart operational skills for chemical plant operations.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Unit operations		Chemical Process Principles	2

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarise size reduction and size separation.	9	Understanding
CO2	Interpret solid - liquid separation equipments.	14	Applying
CO3	Summarise the industrial operations agitation and mixing.	14	Understanding
CO4	Summarise the handling of solids in chemical process industries.	8	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3						
CO3			3				
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarise size reduction and size separation.		
M1.01	Comprehend the properties of particles.	2	Understanding.
M1.02	Summarise size reduction.	1	Understanding.
M1.03	Illustrate size reduction equipment.	3	Understanding.
M1.04	Comprehend the theories behind crushing.	1	Understanding.
M1.05	Summarise the methods of size separation.	2	Understanding.
Contents: Define: Particle shape, particle size, specific surface area of the mixture, average particle size, and shape factor. Size reduction: List the nature of materials to be crushed - List the classification of crushing and grinding equipment. Differentiate closed circuit grinding and open circuit grinding. Illustrate the working of: Blake Jaw crusher - Gyratory crusher -Single Roll crusher and Smooth roll crusher - ball mill and tube mill Explain the theory behind the action of crushing rolls - Define the angle of Nip of crushing rolls. Laws of crushing- State Rittinger's law, Kick's law and Bond's law. Size separation- Compare the Tyler and U.S standard screens. List the types of screening equipment. Illustrate the working of grizzlies, trommel, vibrating screens and magnetic separator. Differentiate shaking and vibrating screen - Ideal and actual screen - Effectiveness of screen			
CO2	Interpret solid - liquid separation equipments.		
M2.01	Comprehend the principles of settling.	4	Understanding
M2.02	Interpret sedimentation processes.	4	Applying
M2.03	Interpret industrial filtration operations.	4	Understanding

M2.04	Comprehend centrifuging operation.	2	Understanding.
	Series Test – I		
Contents: Settling: Types of settling - Stokes law – Define terminal settling velocity. Sedimentation - Batch and continuous sedimentation - Kynch theory – Estimation of thickener area - Illustrate continuous thickener -Illustrate Double cone classifier - Illustrate Tabling, Jigging and Froth floatation. Filtration –Classify filtration equipment based on external force, operation, and nature of filtration - Illustrate the working of sand filter, washing and non-washing plate and frame filter press, leaf filter and Rotary Drum Filter - List the factors affecting rate of filtration - Role of filter aids in filtration - List the commonly used filter aids - Differentiate compressible and non-compressible filter cakes - Differentiate constant pressure filtration and constant volume filtration. Centrifugation: List the classification of centrifuges based on operation and the drive position. Illustrate the working top driven centrifuge, bottom driven centrifuge, disc centrifuge and super centrifuge.			
CO3	Summarise the industrial operations agitation and mixing.		
M3.01	Outline agitation operation.	2	Understanding
M3.02	Summarise agitation equipment.	4	Understanding
M3.03	Identify the dimensionless numbers related to agitation operation.	4	Understanding
M3.04	Summarise the working of mixing equipments.	4	Understanding
Contents: Agitation: Define Agitation - List the purpose of agitation - Flow pattern in agitated vessels. Agitation equipment: Illustrate agitation equipment - Types of Impellers, Sketch of propellers paddles and turbines – Identify the function of draft tube and Baffle. State: Power number, Froude Number and Reynold's no. Mixing: Illustrate dry mixer, Banbury mixer, 'V' type mixer and kneading machine.			
CO4	Summarise the handling of solids in chemical process industries.		
M4.01	Summarise the storage of solids in industries.	2	Understanding
M4.02	Illustrate mechanical conveying systems.	3	Understanding
M4.03	Comprehend pneumatic conveying systems.	3	Understanding
	Series Test – II		

Contents:

Storage of solids- Hoppers, bins, silos - Angle of repose - Open storage of solids.

Classification of conveyors – Illustrate Belt conveyor, Apron conveyor, Screw conveyor, Bucket elevator.

Illustrate Pneumatic conveying system - Illustrate the construction and working of cyclone separator, bag filter and ESP.

Text / Reference

T/R	Book Title/Author
T1	Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero
R1	Mechanical operations; Anup Kumar Swain, hemalathapatra and Gopendra Kishore Roy
R2	Unit Operations of Chemical Engineering; P.Chathopadhyay
R3	Unit Operations; Warren.L.McCabe&Julian.C.Smith,
R4	Perry's Chemical Engineers' Handbook; Robert H Perry, Don W. Green

Online Resources

Sl.No	Website Link
1	https://www.nptel.ac.in/courses/103/107/103107123/